D 130229

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Name.....

Reg. No.....

Fifth Semester (CBCSS—UG) Degree Examination, November 2025

Mathematics

MTS 5B 08—Linear Programming

(2020 Syllabus)

Time : Two Hours Maximum : 60 Marks

Section A (Short Answer Type)

All questions can be answered. Each question carries 2 marks. (Ceiling 20 marks)

1 The equation of the line segment joining $(2, 1)$ and $(5, 7)$ is ————————.

2 Define a hyperplane in $\mathbb{R}^2$.

3 Convert the linear programming problems given below to canonical form :
Maximize $f(x, y) = -x - 2y$ subject to $2x - y \geq 10$, $3y - x \leq 8$, $x \geq 0$, $y \geq 0$.

4 Draw and shade a convex subset of $\mathbb{R}^2$ having no extreme points.

5 Consider the canonical maximum tableau below :

```
      x     y    -1
 3    1     5    = -t1
-7    1     8    = -t2
 4    9    13    =  f
```

State the canonical maximization linear programming problem represented by the tableau above.

Turn over

6 Write the basic solution of the linear programming problem given below :
Maximize $f(x, y) = 200x + 150y$ subject to $x + 2y \leq 11$, $2x + 2y \leq 30$, $2x + 2y \leq 25$, $x \geq 0$, $y \geq 0$.

7 Write the canonical slack minimization linear programming problem.

8 State Von-Neumann Minimax Theorem.

9 What is complementary slackness of a dual canonical linear programming problem ?

10 What is balanced transportation problem ?

11 What is the pure strategy of a matrix game ?

12 What is a basic feasible solution of a transportation problem ?

Section B (Paragraph/Problem Type)

All questions can be answered. Each question carries 5 marks. (Ceiling 30 marks)

13 Show that the linear programming problem :
Maximize $f(x, y) = 3x + 2y$ subject to $2x - y \leq -1$, $x - 2y \geq 0$, $x \geq 0$, $y \geq 0$.
is infeasible.

14 Solve the linear programming problem graphically :
Maximize $f(x, y) = 5x + 2y$ subject to $x + 3y \leq 14$, $2x + y \leq 8$, $x \geq 0$, $y \geq 0$.

15 Apply the simplex algorithm to the maximum tableau given below :

```
      x1   x2   -1
 2     1    8   = -t1
 1     2   10   = -t2
30    50   10   =  f
```

16 Solve the unconstrained problem :
Maximize $f(x, y) = x + 3y$ subject to $x + 2y \leq 10$, $-3x - y \leq -15$.

17 A tableau of the balanced assignment problem is given below :

1	0	0
0	1	0
1	0	1

Find a permutation set of zeros.

18 Solve the assignment problem given below :

38	21	34
41	14	36
28	20	25

19 What is a two-person zero-sum matrix game ?

Section C (Essay Type)

Answer any one of the following question. The question carries 10 marks.

20 Solve the transportation problem by using Vogel Advanced-Start Method :

	M1	M2	M3	
W1	2	1	2	40
W2	9	4	7	60
W3	1	2	9	10
	40	50	20	110

21 A company wishes to assign five of its workers to five different jobs. The rating of each worker with respect to each job on a scale of 0 to 10 is given by the following table :

	J1	J2	J3	J4	J5
W1	5	4	2	8	5
W2	7	6	4	6	9
W3	5	5	3	3	2
W4	4	3	5	5	4
W5	3	6	4	10	2

If the company wishes to maximize the total rating of the assignment, find the optimal assignment plan and the corresponding maximum total rating.

(1 × 10 = 10 marks)